

Ain Shams University

International Credit Hours

Faculty of Engineering



Transylvania Hotel

Hotel Reservation Management System

Project Report

Object-Oriented Programming —Project milestone 2
Spring 2026

Team Members

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1. Introduction

The Transylvania Hotel Reservation Management System is a desktop application built using Java and JavaFX. It provides a complete hotel management solution covering guest registration, room browsing, reservation management, billing, and real-time staff coordination. The system is backed by a local SQLite database and features a clean, themed UI.

2. System Overview

2.1 Architecture

The system follows:

- Models: Guest, Staff, Admin, Receptionist, Room, RoomType, Reservation, Invoice, Amenity
- Views: FXML files (login.fxml, Register.fxml, Dashboard.fxml, staff_dashboard.fxml, RoomBrowsing.fxml, Checkout.fxml)
- Controllers: One controller per view, handling all user interactions and DB persistence
- Database Layer: HotelDatabase.java — single source of truth, loads all data into in-memory ArrayLists on startup
- Live chats feature

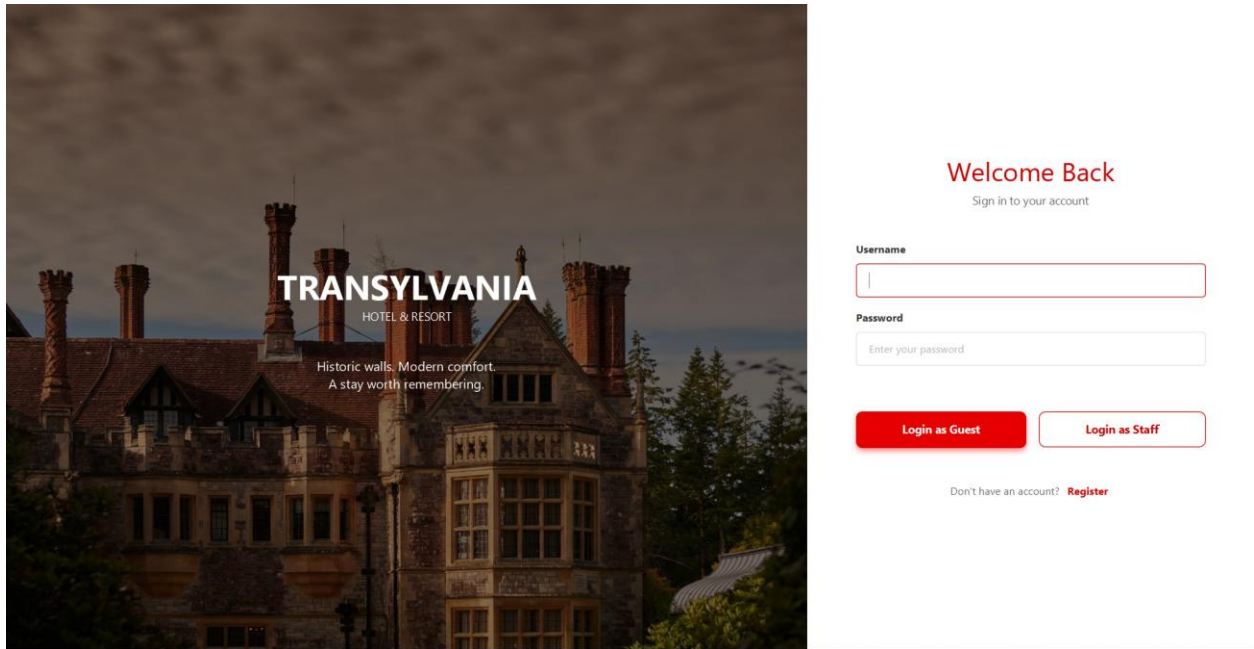
2.2 OOP Principles Applied

- Encapsulation: All model fields are private with validated getters/setters (e.g. password min 7 chars, balance cannot be negative)
- Inheritance: Admin and Receptionist extend Staff, which extends Person. Guest also extends Person
- Polymorphism: Staff roles (Admin/Receptionist) share common interfaces but behave differently — Admin manages rooms, Receptionist handles check-ins
- Abstraction: Payable interface implemented by Guest; Person abstract class defines common identity fields
- Exception Handling: Custom exceptions — RoomNotAvailableException, InvalidDateRangeException, InvalidPaymentException

3. Features & Modules

3.1 Authentication — Login & Register

Both the login and registration screens feature a two-panel layout. The background image is loaded from the classpath (src/images/back1.jpg) so it works on any machine that clones the repository.



The login screen features a two-panel layout. The left panel is a large image of a historic stone building with multiple chimneys and a bay window, set against a dark, cloudy sky. Overlaid on the image is the text "TRANSYLVANIA HOTEL & RESORT" in white, with the tagline "Historic walls. Modern comfort. A stay worth remembering." below it. The right panel is a white form with a red header "Welcome Back" and a sub-header "Sign in to your account". It contains fields for "Username" and "Password", with a placeholder "Enter your password" in the password field. Below the fields are two red buttons: "Login as Guest" and "Login as Staff". At the bottom, there is a link "Don't have an account? Register" in red.

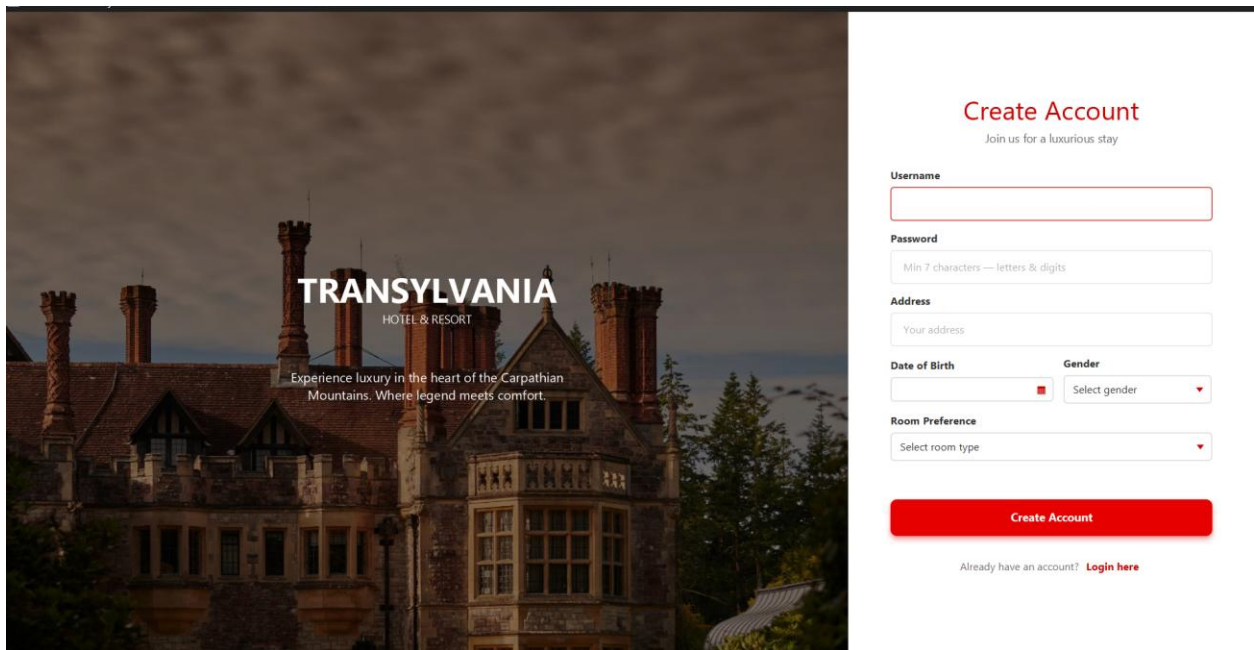
Welcome Back
Sign in to your account

Username

Password

[Login as Guest](#) [Login as Staff](#)

Don't have an account? [Register](#)



The registration screen features a two-panel layout. The left panel is a large image of the same historic stone building as the login screen, with the text "TRANSYLVANIA HOTEL & RESORT" and the tagline "Experience luxury in the heart of the Carpathian Mountains. Where legend meets comfort." overlaid. The right panel is a white form with a red header "Create Account" and a sub-header "Join us for a luxurious stay". It contains fields for "Username", "Password", "Address", "Date of Birth", "Gender", and "Room Preference". The "Password" field has a placeholder "Min 7 characters — letters & digits". The "Date of Birth" field has a red square icon. The "Gender" field has a dropdown menu with "Select gender". The "Room Preference" field has a dropdown menu with "Select room type". Below the fields is a red button "Create Account". At the bottom, there is a link "Already have an account? Login here" in red.

Create Account
Join us for a luxurious stay

Username

Password

Address

Date of Birth Gender

Room Preference

[Create Account](#)

Already have an account? [Login here](#)

3.2 Guest Dashboard

After logging in, guests see a personalized dashboard showing their balance, room preference, and list of active reservations. From here they can:

- Browse available rooms and make a reservation
- Pay their bill through the Checkout screen
- Cancel an active reservation (with automatic refund if already paid)
- Log out

TRANSYLVANIA HOTEL & RESORTGuest PortalChatLogout

Welcome, Mariam!
Manage your reservations and explore our rooms

Account Balance
Balance: \$15000.0

Active Reservations
0

Room Preference
Room Preference: Suite

Your Reservations
No reservations yet.

Quick Actions
[Browse Rooms](#)
[Checkout and Pay](#)
[Cancel Reservation](#)
[Chat with Receptionist](#)
Need Help?
Contact our reception desk or use the live chat feature to speak with staff.

3.3 Room Browsing & Reservation

Guests can browse all available rooms with photos, room type, floor, and price per night. Room images are loaded from the classpath (src/images/) Selecting a room and choosing check-in/check-out dates creates a reservation with status PENDING.

[< Back](#) **Browse Available Rooms**

Filters
Room Type
All Types
Price Range
All Prices
Floor
All Floors

Available Rooms
9 rooms available
Sort by: Price: Low to High

Room 102 ★ 4.5
Double
\$90 per night
Floor 1 2 guests 32 m²
Amenities: WiFi TV AC

Room 202 ★ 4.5
Double
\$90 per night
Floor 2 2 guests 32 m²
Amenities: WiFi AC

Room 203 ★ 4.5
Double
\$90 per night
Floor 2 2 guests 32 m²
Amenities: WiFi TV AC

Room 301 ★ 4.5

Room 505 ★ 4.5

Room 302 ★ 4.5

[Book Selected Room](#)
[Cancel](#)

9 rooms available | No selection

Transylvania Hotel Management System v1.0

3.4 Checkout & Payment

When a guest is ready to pay, they are taken to the Checkout screen which shows the total amount due (nights × price per night). The guest selects a payment method (CASH, CREDIT_CARD, ONLINE) and pays the exact amount. A paid invoice is recorded. The reservation status remains CONFIRMED and the room stays Occupied — the room only becomes Available again after the receptionist physically checks the guest out.

TRANSYLVANIA HOTEL & RESORT

Make a ReservationBack

Reservation Details

Review your room and choose your dates

Selected Room

Room No.

Room: 202

Type

Type: Double

Price / Night

Price: \$90.0 / night

Floor

Floor: 2

Amenities

Amenities: WiFi, AC

Choose Your Dates

Check-in Date

Check-out Date

Calculate Total

Confirm Reservation

Choose Your Dates

Check-in Date

5/6/2026

Check-out Date

5/14/2026

Unexpected error: Check-in date cannot be in the past.

Choose Your Dates

Check-in Date

5/9/2026

Check-out Date

5/7/2026

Check-out must be after check-in.

TRANSYLVANIA HOTEL & RESORT

Make a ReservationBack

Reservation Details

Review your room and choose your dates

Selected Room

Room No.

Room: 202

Type

Type: Double

Price / Night

Price: \$90.0 / night

Floor

Floor: 2

Amenities

Amenities: WiFi, AC

Choose Your Dates

Check-in Date

5/9/2026

Check-out Date

5/14/2026

Success

!

Dates confirmed! Please proceed to payment.

OK

Calculate Total

Nights: 5 | Total: \$450.0

Confirm Reservation

TRANSYLVANIA HOTEL & RESORT

CheckoutDashboard

Payment Details

Complete your checkout below

Amount to Pay

450

Payment Method

CREDIT_CARD

Credit Card

Cash

Online Transfer

Pay Now

AMOUNT DUE

\$450.00

Transylvania Hotel and Resort

Secure payment processing

TRANSYLVANIA HOTEL & RESORT

CheckoutDashboard

AMOUNT DUE

\$450.00

Transylvania Hotel and Resort

Secure payment processing

Payment Details

Complete your checkout below

Amount to Pay

450

Payment Method

CREDIT_CARD

Processing payment...

Payment Successful

Payment successfull The room is now reserved and fully paid.

OK

Credit Card

Cash

Online Transfer

Pay Now

TRANSYLVANIA HOTEL & RESORT

Guest PortalChatLogout

Welcome, Mariam!

Manage your reservations and explore our rooms

Account Balance

Balance: \$14550.0

Active Reservations

1

Room Preference

Room Preference: Suite

Your Reservations

Room 202 | 2026-05-09 – 2026-05-14 | PENDING | Paid

Quick Actions

Browse Rooms

Checkout and Pay

Cancel Reservation

Chat with Receptionist

Need Help?
Contact our reception desk or use the live chat feature to speak with staff.

3.5 Staff Dashboard — Admin

Admin staff have access to all three tabs: Guests, Rooms, and Reservations. The left sidebar shows live counters for total Guests, total Rooms, and Occupied rooms. The active nav button is highlighted in red to show the current tab.

- Guests tab: View all registered guests with email, phone, balance, and room preference
- Rooms tab: View all rooms with type, floor, price, and availability status. Admins can create new rooms, update a room's type (price updates automatically), and delete rooms
- Reservations tab: View all reservations

Transylvania Hotel - Staff Portal

Role: ADMIN
Dashboard - omar

Chat

Logout

Staff User
System Operator

5
Guests

9
Rooms

1
Occupied

Guests

Rooms

Reservations

Guests Management

Manage and view registered guests

GUEST	EMAIL	PHONE	BALANCE	PREFERENCE
 Mariam	mariam@email.com	+1 234 567 -3899	\$14550.00	Suite
 Ibrahim	ibrahim@email.com	+1 234 567 -6894	\$700.00	Single
 Carol	carol@email.com	+1 234 567 5902	\$1000.00	Suite
 Sara	sara@email.com	+1 234 567 5204	\$2000.00	Double
 Ahmed	ahmed@email.com	+1 234 567 9091	\$1500.00	Single

System Ready

Transylvania Hotel Management System v1.0 | Staff Mode

Transylvania Hotel - Staff Portal

5

9

1

Guests

Rooms

Occupied

Guests

Rooms

Reservations

Staff User

System Operator

Rooms Management

Manage room inventory and prices

+ Create Room

Update Selected

Delete Selected

ROOM	TYPE	FLOOR	PRICE	STATUS
Room 101	Single	Floor 1	\$50.00	Available
Room 102	Double	Floor 1	\$90.00	Available
Room 505	Suite	Floor 2	\$200.00	Available
Room 202	Double	Floor 2	\$90.00	Occupied
Room 103	Single	Floor 1	\$50.00	Available
Room 104	Single	Floor 1	\$50.00	Available
Room 203	Double	Floor 2	\$90.00	Available
Room 301	Suite	Floor 3	\$150.00	Available
Room 302	Suite	Floor 3	\$200.00	Available

System Ready

Transylvania Hotel Management System v1.0 | Staff Mode

Transylvania Hotel - Staff Portal

5

9

1

Guests

Rooms

Occupied

Guests

Rooms

Reservations

Staff User

System Operator

Reservations Management

Handle check-ins and check-outs

GUEST	ROOM	CHECK-IN	CHECK-OUT	STATUS
Mariam	Room 202	2026-05-09	2026-05-14	PENDING

System Ready

Transylvania Hotel Management System v1.0 | Staff Mode

3.6 Staff Dashboard — Receptionist

Receptionists see the Reservations tab with Check-In and Check-Out actions. The correct reservation status flow is:

PENDING Guest booked	CONFIRMED Receptionist checked in	COMPLETED Guest checked out	CANCELLED Reservation cancelled
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Transylvania Hotel - Staff Portal

Role: RECEPTIONIST

Dashboard - aly

Chat

Logout

Staff User

System Operator

5

Guests

9

Rooms

1

Occupied

Guests

Rooms

Reservations

Reservations Management

Handle check-ins and check-outs

Check-In Selected

Check-Out Selected

GUEST	ROOM	CHECK-IN	CHECK-OUT	STATUS
Mariam	Room 202	2026-05-09	2026-05-14	PENDING

Transylvania Hotel - Staff Portal

Role: RECEPTIONIST

Dashboard - aly

Chat

Logout

Staff User

System Operator

5

Guests

9

Rooms

1

Occupied

Guests

Rooms

Reservations

Handled Check-in for: Mariam

Reservations Management

Handle check-ins and check-outs

Check-In Selected

Check-Out Selected

GUEST	ROOM	CHECK-IN	CHECK-OUT	STATUS
Mariam	Room 202	2026-05-09	2026-05-14	CONFIRMED

Transylvania Hotel - Staff Portal

Role: RECEPTIONIST

Dashboard - aly

Chat

Logout

Staff User

System Operator

5

Guests

9

Rooms

0

Occupied

Guests

Rooms

Reservations

Handled Check-out for: Mariam

Reservations Management

Handle check-ins and check-outs

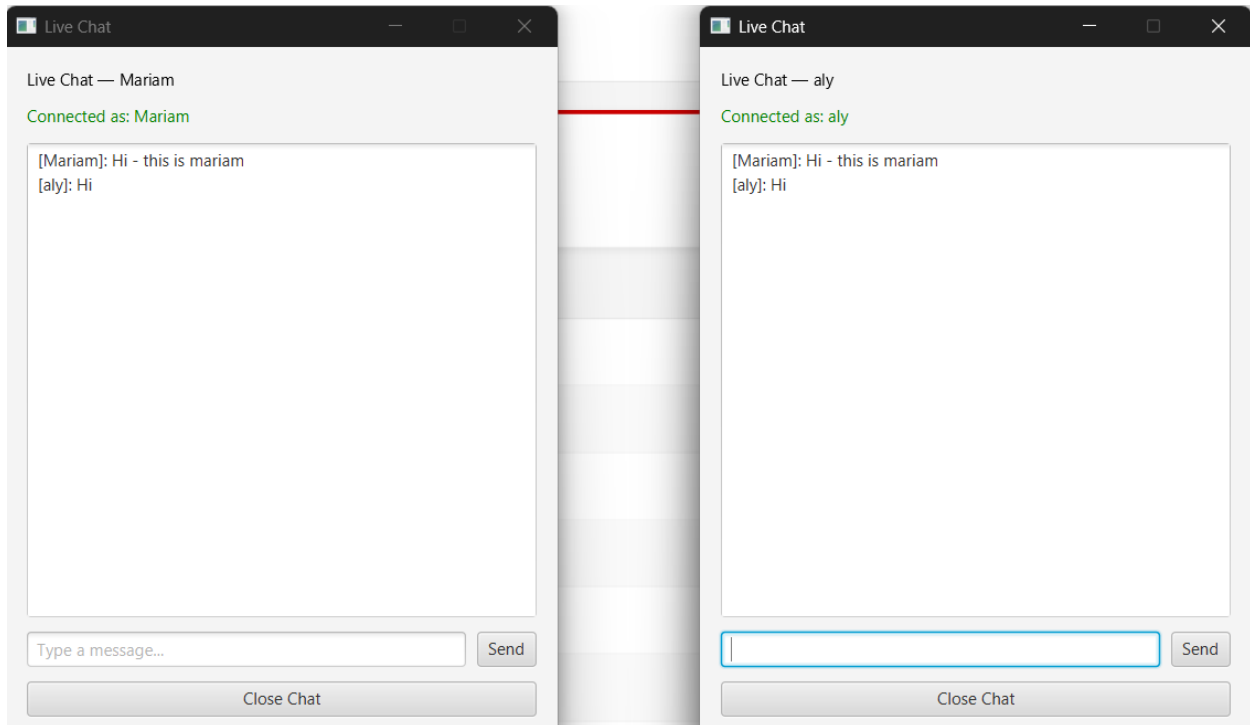
Check-In Selected

Check-Out Selected

GUEST	ROOM	CHECK-IN	CHECK-OUT	STATUS
Mariam	Room 202	2026-05-09	2026-05-14	COMPLETED

3.7 Chat System

The system includes a built-in real-time chat feature. A ChatServer runs on port 5001 when the app starts. Guests and staff can open the chat window to communicate in real time using Java sockets and multithreading.



4. Database Design

The system uses SQLite with the following tables:

Table	Key Columns
guests	username (PK), password, dateOfBirth, address, gender, balance, roomPreferences, phone
staff	username (PK), password, dateOfBirth, address, gender, role (ADMIN/RECEPTIONIST), workingHours
rooms	roomId (PK), roomNumber, floor, isAvailable (0/1), pricePerNight, roomTypeId (FK)
room_types	typeId (PK), typeName (Single/Double/Suite), basePrice
reservations	id (PK), guestUsername (FK), roomId (FK), checkInDate, checkOutDate, status
invoices	id (PK), reservationId (FK), amount, paymentMethod, paid (0/1)
amenities	amenityId (PK), name, description
room_amenities	roomId (FK) + amenityId (FK)

On first launch, if no data exists, the system seeds the database with dummy data: room types, amenities, rooms with amenities, 5 guests, and 4 staff members (2 admins, 2 receptionists). This ensures anyone who clones the repo gets a working demo immediately.

5. Test Cases

The following test cases cover both valid and invalid scenarios across all major modules.

5.1 Authentication

Description	Input	Expected	Result	Status
Valid guest login	username: Mariam, password: Mario123	Redirected to Guest Dashboard	Guest Dashboard loads successfully	PASS
Valid staff login (Admin)	username: omar, password: omar1234	Redirected to Staff Dashboard (ADMIN role)	Staff Dashboard loads with Admin privileges	PASS
Valid staff login (Receptionist)	username: aly, password: aly12345	Redirected to Staff Dashboard (RECEPTIONIST role)	Staff Dashboard loads with Receptionist privileges	PASS
Invalid password	username: Mariam, password: wrongpass	Error message shown, no navigation	Error displayed: invalid credentials	PASS
Empty login fields	username: (empty), password: (empty)	Error message, login blocked	Validation error shown	PASS
Register with short password	password: abc123 (6 chars)	Error: password must be at least 7 chars	Registration blocked with error message	PASS
Register with duplicate username	username already exists in DB	Error: username taken	Registration blocked with error message	PASS
Valid new guest registration	All fields filled correctly	Account created, redirected to login	Guest account created in DB	PASS



TRANSYLVANIA

HOTEL & RESORT

Historic walls. Modern comfort.
A stay worth remembering.

Welcome Back

Sign in to your account

Username

Mariam

Password

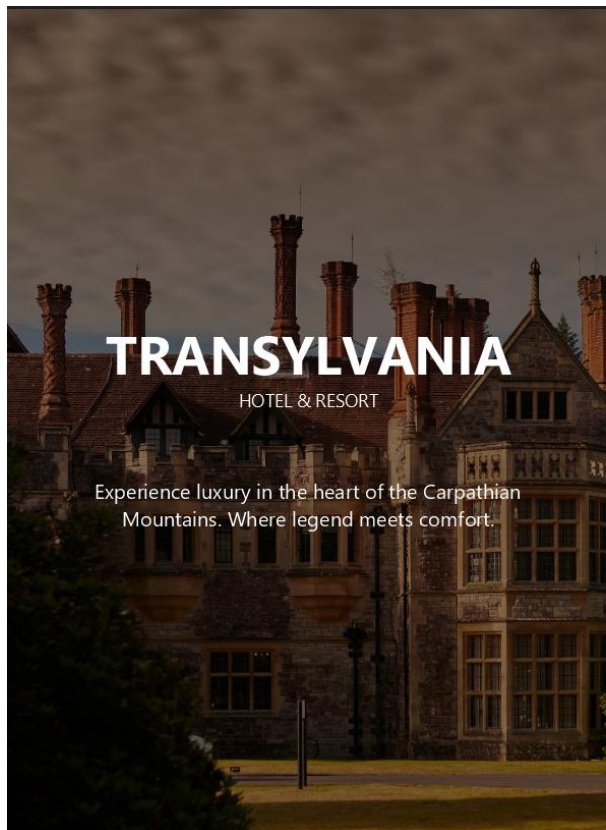
••••••

Invalid username or password.

Login as Guest

Login as Staff

Don't have an account? [Register](#)



TRANSYLVANIA

HOTEL & RESORT

Experience luxury in the heart of the Carpathian
Mountains. Where legend meets comfort.

Create Account

Join us for a luxurious stay

Username

Min 3 characters

Password

••••

Address

Your address

Date of Birth

Gender

Select gender

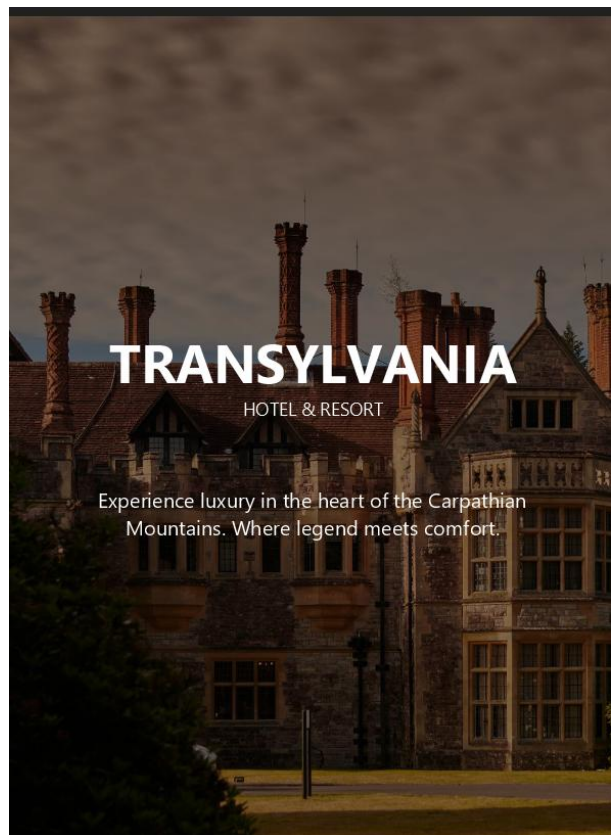
Room Preference

Select room type

Please fill in all fields.

Create Account

Already have an account? [Login here](#)



TRANSYLVANIA

HOTEL & RESORT

Experience luxury in the heart of the Carpathian Mountains. Where legend meets comfort.

Create Account

Join us for a luxurious stay

Username

werw

Password

••••

Address

werw

Date of Birth

4/28/2026

Gender

FEMALE

Room Preference

Double

Error: Password must be at least 7 characters long.

Create Account

Already have an account? [Login here](#)

5.2 Room Browsing & Reservation

Description	Input	Expected	Result	Status
Browse available rooms	Guest opens Room Browsing	All available rooms shown with images and prices	Rooms displayed with correct type images	PASS
Reserve an available room	Select room, valid check-in/out dates	Reservation created with PENDING status	Reservation visible in Guest Dashboard	PASS
Reserve already-booked room	Select occupied room	Error: room not available	Booking blocked with error	PASS
Check-in date in the past	check-in: yesterday's date	Error: cannot book past dates	Date validation error shown	PASS
Check-out before check-in	check-out earlier than check-in	Error: invalid date range	Date range error shown	PASS
Overlapping reservation dates	Dates overlap with existing booking	Error: room already booked for these dates	Overlap detection works	PASS

5.3 Payment & Checkout

Description	Input	Expected	Result	Status
Pay exact required amount	Correct amount entered, method selected	Invoice created, balance deducted, room stays Occupied	Payment successful, returns to dashboard	PASS
Pay wrong amount	Amount less than or more than required	Error: must pay exact amount	Payment blocked with error	PASS
Pay with insufficient balance	Guest balance less than required	Error: insufficient balance	Payment blocked	PASS
No payment method selected	Amount correct, method not chosen	Error: select payment method	Validation error shown	PASS
Cancel reservation with refund	Cancel a paid reservation	Balance refunded, reservation CANCELLED, room Available	Refund processed correctly	PASS

5.4 Staff — Rooms Management (Admin)

Description	Input	Expected	Result	Status
Create a new room	Fill in room number, floor, type	Room added to DB and list	New room appears in Rooms tab	PASS
Update room type	Change Single room to Suite	Price updates to Suite base price (\$200)	Price correctly updates in table	PASS
Delete a room	Select room and click Delete	Room removed from DB	Room disappears from list	PASS
Occupied counter accuracy	Rooms with isAvailable=false	Counter shows correct occupied count	Counter matches actual occupied rooms	PASS

5.5 Staff — Reservation Flow (Receptionist)

Description	Input	Expected	Result	Status
Check-in a PENDING reservation	Select PENDING reservation, click Check-In	Status → CONFIRMED, persisted to DB	Status updates in table	PASS
Check-in an already CONFIRMED reservation	Select CONFIRMED reservation, click Check-In	Error: already confirmed	Error message shown	PASS
Check-out a CONFIRMED reservation	Select CONFIRMED reservation, click Check-Out	Status → COMPLETED, room → Available	Room status updates, counter decreases	PASS
Check-out a PENDING reservation	Select PENDING reservation, click Check-Out	Error: must be CONFIRMED first	Error message shown	PASS
Check-in before check-in date	Reservation check-in date is in the future	Error: check-in date not reached	Early check-in blocked	PASS